Technical Specification

The proposed architecture for the NFT digital twin is shown in Figure 1 below. The hardware will be a consortium EVM blockchain, wherein a small number of trusted players are involved. The focus is on traceable provenance/origination, easy audits, and immutability (allowing it to be tamper-proof).
Each validator would run the execution client to store and updates states and a validator service is used to produce and finalise blocks and allow for API access

Node/network roles:
Validators: this would be a small consortium set ideally including CNES or similar public bodies (see design motivation for further details), prime contractors, authorised distributors and testing labs
Non-validator members: manufacturers, auditors
Wallets/users: used by supply-chain actors to mint, transfer, split etc
API servers: interact with contract events and ledger to organise and store information for dashboards/analytics

Network Topology
Mesh network as this is more reliable than line or ring structures
faster than a public network
good for small known number of members
one or two nodes can fail without issues

Data Storage and Setup
See figure 2

Example API calls
These are roughly linked to data model 1 and data model 2

1) Provenance Lookup
This is designed to reconsruct the history of a part lot from minting to its end destination, helping with traceability in mass production.

GET /api/provenance/10

On-chain interactions:
ownerOf(10)
twins(10)
which retrives
TwinMinted, CustodyTransferred, ReelSplit, StatusUpdated

Off-chain interactions:
document metadata storage, records linked through URIs
Example response:

```
{
  "tokenId": 10,
  "partNumber": "ABC-123",
  "manufacturer": "0x123...",
  "owner": "0x456...",
```

```
  "quantity": 1200,
  "parentTokenId": 11,
  "status": "Active",
  "documents": [
    {
      "docHash": "0xaaa...",
      "uri": "...",
      "type": "radiation_test_report",

    }
  ],
  "custodyTrail": [
    {
      "from": "0x123...",
      "to": "0x456...",
      "event": "CustodyTransferred"
    }
  ]
}
```


2) Documentation Assessment

This API call is provides an operational assessment of a particular lot. This helps with both lineage and also documentation scarcity.

GET /api/documentation-score /10

<u>On-chain interactions:</u>
getDocumentCount(10)
getDocument(10)
twins(10).status
origination trail from parentTokenId

<u>Off-chain interactions:</u>
document metadata storage, lab certification data, supplier database
<u>Example response:</u>


```
{
  "tokenId": 10,
  "score": 82,
  "result": "review recommended",
  "checks": {
    "hashVerified": true,
    "lineageComplete": true,
    "testReportExists": true,
    "authorizedDistributorEvidence": False,
  }
}
```

Figure 1: Architectural Diagram

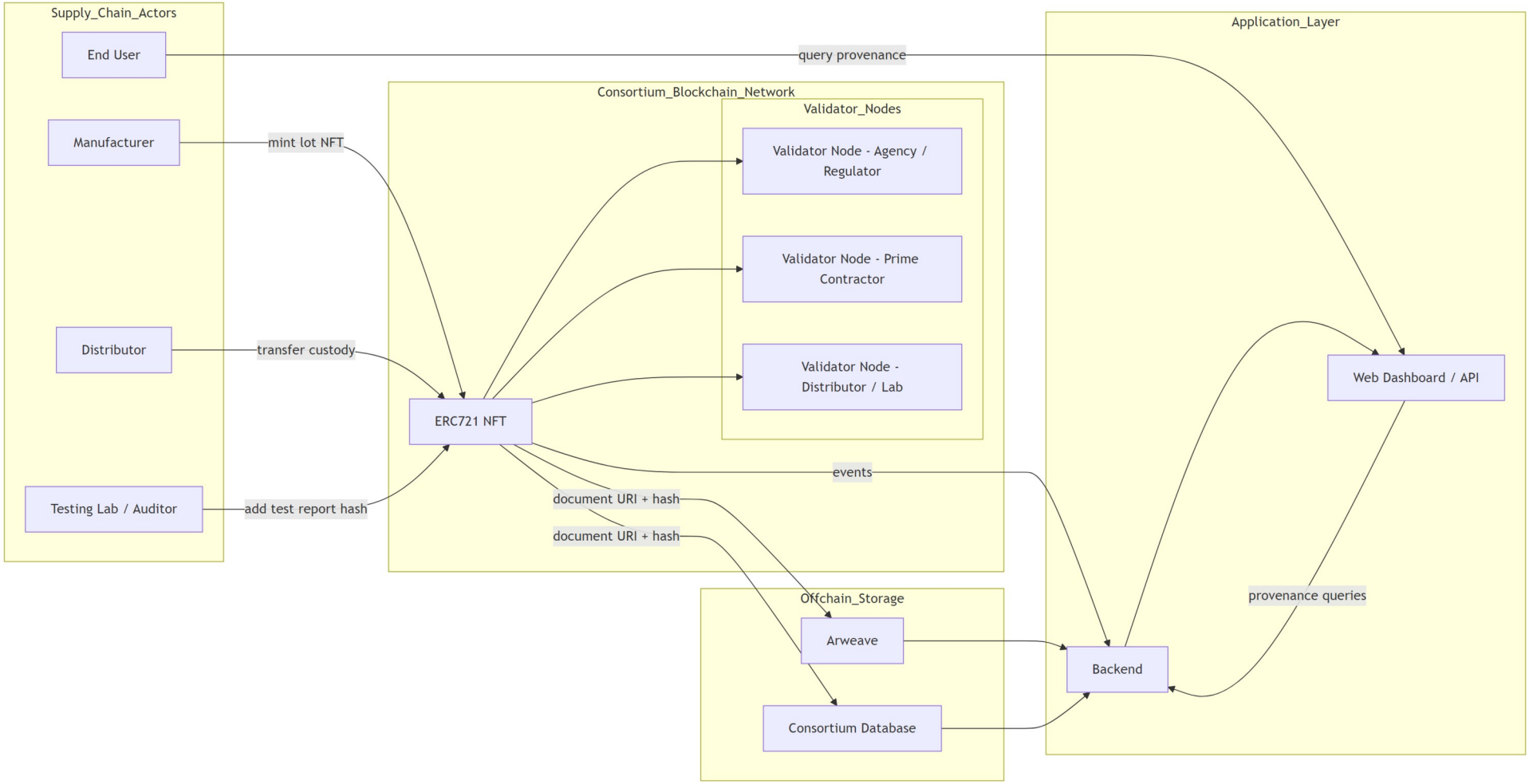

Figure 2: Offline vs Online Data

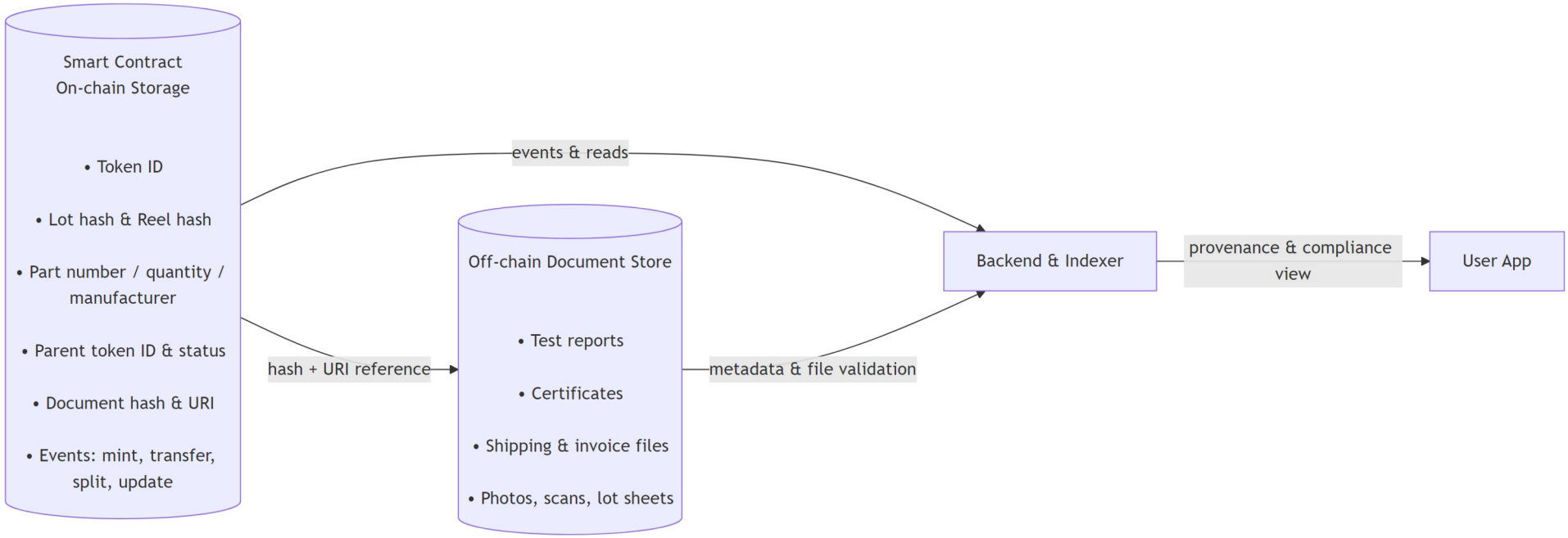

Figure 3: Data Model 1 - Provenance

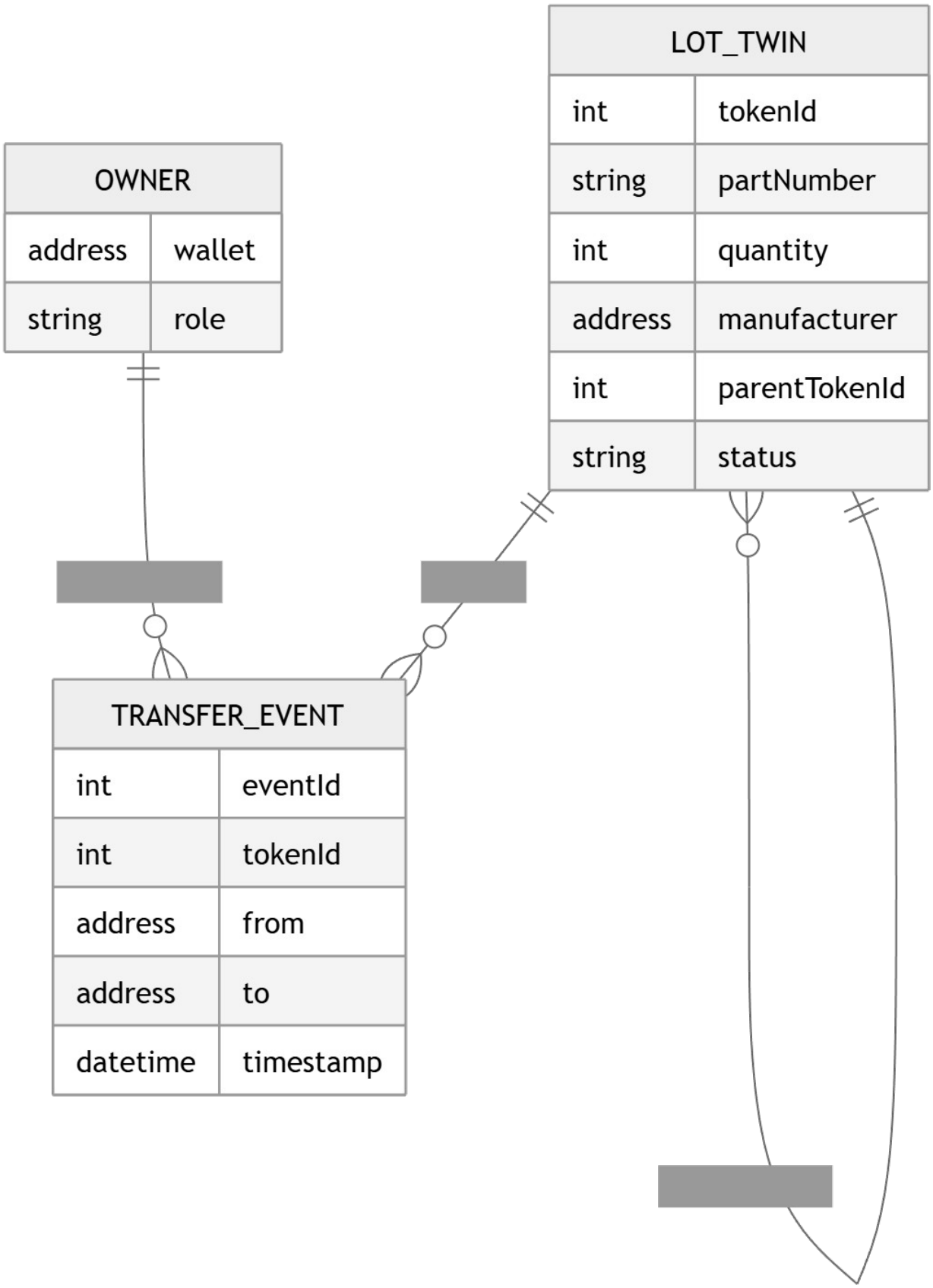
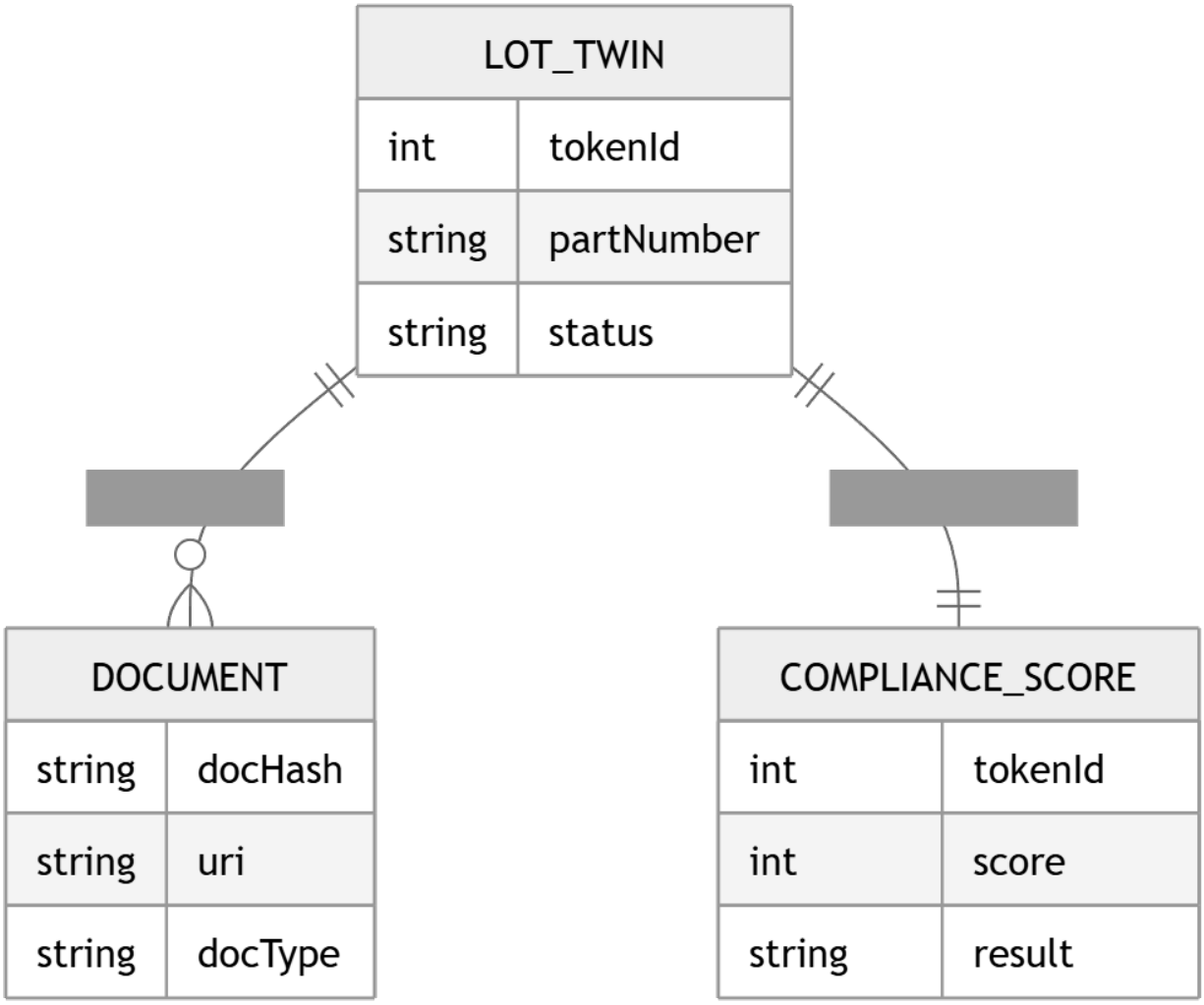
Figure 4: Data Model 2 - Documentation